

AI Model Card / Transparency

Grounded in: LumineticsCore™ / IDx-DR (De Novo DEN180001)

Basis: FDA AI transparency principles; GMLP.

Device	LumineticsCore™ / IDx-DR — autonomous diabetic retinopathy AI
Status	De Novo DEN180001 (2018); Class II; two algorithms (image-quality + diagnostic)
Basis	FDA AI transparency principles; GMLP; EU AI Act. Illustrative.

Intended use & users

Autonomous detection of more-than-mild diabetic retinopathy (mtmDR) in adults with diabetes, in primary care, without a clinician interpreting the image.

Inputs / outputs

Fundus images from specified cameras → image-quality gate → mtmDR detected / not detected.

Performance (public pivotal trial)

Metric	Value
FDA performance threshold — sensitivity	85%
FDA performance threshold — specificity	82.5%
Pivotal study	900 patients, primary-care sites, independent reference standard

Limitations

- Not for patients previously diagnosed with DR.
- Image-quality dependent (built-in gate).
- Specified cameras only.

Monitoring & change control

Real-world performance monitoring; retraining governed by a Predetermined Change Control Plan.

Sources (public)

- FDA De Novo DEN180001 (IDx-DR), 2018; Digital Diagnostics (LumineticsCore) public materials.
- Abràmoff et al., npj Digital Medicine (2018) — pivotal trial; thresholds 85% sensitivity / 82.5% specificity.
- FDA — Marketing Submission Recommendations for a PCCP for AI-Enabled Device Software Functions (final, 2024).